

1 K-Nearest Neighbour Regression

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[1]: # import required libraries
import pandas as pd
import math

data = pd.read_csv("data.csv")
data
```

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[1]:      Height  Weight  Target
0       150      50     1.5
1       155      55     1.2
2       160      60     1.8
3       161      59     2.1
4       158      65     1.7
```

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[2]: # run the code
def euclidean_distance(p1, p2):
    return math.sqrt(sum((a - b) ** 2 for a, b in zip(p1, p2)))
```

```
[3]: # run the code
def knn_regression(X_train, y_train, X_test, k):
    distances = []

    for i in range(len(X_train)):
        distances.append((float(y_train[i]), euclidean_distance(X_test, X_train[i])))

    print(distances)
    distances.sort(key=lambda x: x[1])
    mean = 0

    for target, _ in distances[:k]:
        mean += target

    return mean / k
```

```
[4]: # prepare the data
X_train = data.iloc[:, :-1].values
y_train = data.iloc[:, -1].values

# Step 6: Example test point
X_test = [157, 54]

print(X_test)
k = 3
y_pred = knn_regression(X_train, y_train, X_test, k)

[157, 54]
[(1.5, 8.06225774829855), (1.2, 2.23606797749979), (1.8, 6.708203932499369),
(2.1, 6.4031242374328485), (1.7, 11.045361017187261)]
```

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[5]: # display the result
print(y_pred)
```

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